

Fluid Dynamics of the Atmosphere and Ocean

Problem Set 4

This problem set counts 10% toward your final mark. Due on March 20.

In all cases partial credit will be given if the algebra is wrong but the method right, *provided you explain what you are trying to do.*

1. Geostrophic adjustment with a velocity jump.

- (a) Show that the *linearized* potential vorticity, q' , for the shallow water system is given by

$$q' = \zeta' - f_0 \frac{h'}{H},$$

using standard notation.

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- (b) If the flow is in geostrophic balance show that the relative vorticity is given by

$$\zeta' = \nabla^2 \psi.$$

where $\psi = gh'/f_0$. Hence show that the potential vorticity is then given by

$$q' = \nabla^2 \psi - \frac{1}{L_d^2} \psi,$$

and write down an expression for L_d . (If you wish you may take $g = 1$.)

{5}

- (c) Suppose that the initial flow has a flat surface with $u' = 0$ everywhere and Suppose that initially the fluid surface is flat, the zonal velocity (u) is zero and the meridional velocity is given by

$$v(x) = v_0 \operatorname{sgn}(x)$$

- (i) Find the equilibrium height and velocity fields at $t = \infty$, in the linear approximation.

- (ii) What are the initial and final kinetic and potential energies?

Hint: The potential vorticity is $q = \zeta - f_0 \eta/H$, so that the initial potential vorticity is given by

$$q = 2v_0 \delta(x).$$

where $\delta(x)$ is the Dirac delta function.

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2. Vertically propagating Rossby waves

- (a) The quasigeostrophic potential vorticity equation, linearized about a constant eastward mean flow U is

$$\left(\frac{\partial}{\partial t} + U \frac{\partial}{\partial x} \right) q + \beta v = 0,$$

where

$$q = \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{1}{\rho_0} \frac{\partial}{\partial z} \left(\rho_0 \frac{f_0^2}{N^2} \frac{\partial \psi}{\partial z} \right)$$

is the perturbation potential vorticity. Here, ψ is the perturbation geostrophic stream function, z is a vertical coordinate, and ρ_0 , f_0 , N_{ref} and $\beta > 0$ are constants.

By seeking solutions of the form

$$\psi = \text{Re} \{ \hat{\psi} \exp [i (kx + ly + mz - \omega t)] \},$$

where k and l are real, derive the dispersion relation

$$\omega = kU - \frac{k\beta}{k^2 + l^2 + m^2 (f_0^2/N^2)}.$$

What kind of waves satisfy this dispersion relation?

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- (b) Calculate the vertical component of the group velocity for such waves. Hence, sketch the orientation in the $x - z$ plane of the crests and troughs required for upward propagation of a packet of these waves.

{4}

- (c) The perturbation v and θ components will also have wavelike behaviour

$$v = \text{Re} \{ \hat{v} \exp [i (kx + ly + mz - \omega t)] \},$$

$$\theta = \text{Re} \{ \hat{\theta} \exp [i (kx + ly + mz - \omega t)] \}.$$

where $g\theta/\theta_0 = f_0\partial\psi/\partial z$. The northward eddy potential temperature flux is defined as

$$\rho_0 \overline{v\theta} \equiv \rho_0 \left(\frac{k}{2\pi} \right) \int_{\text{one wavelength}} v\theta dx = \frac{1}{2} \rho_0 \text{Re} \{ \hat{v} \hat{\theta}^* \},$$

where $\hat{\theta}^*$ indicates the complex conjugate of $\hat{\theta}$. By expressing $\hat{v}$ and $\hat{\theta}$ in terms of $\hat{\psi}$, and assuming k and m are real, show that

$$\overline{v\theta} = \frac{1}{2} \theta_0 k m \frac{f_0}{g} |\hat{\psi}|^2.$$

Hence explain why upward wave propagation requires a poleward eddy potential temperature flux.

{5}

- (d) Now consider *stationary* waves, and suppose that k and l are given and real (and, as before, $\beta > 0$). Show that upward wave propagation is possible only if U satisfies the inequality

$$\frac{\beta}{k^2 + l^2} > U > 0.$$

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3. Eastward jets

1. Show that there can be no extrema of a materially conserved tracer in the interior of a fluid in the presence of a diffusion acting on the tracer (that is, a tracer obeying $D\phi/Dt = \kappa \nabla^2 \phi$ with $\kappa > 0$), if the flow is steady. If you wish, and without loss of credit, you may restrict your answer to incompressible flows. {6}
2. Show that the results of part (a) imply that there can be no steady (non-eddy) mean eastward flow at the equator provided that the vertical diffusion of zonal momentum dominates the horizontal diffusion. State and explain carefully any assumptions you make. {6}
3. Show that if there is a source of Rossby waves at any given latitude, we expect that eastward flow will be generated there. {8}

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